The generator approach of Stein's method

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Orsay - December 2024





Plan

Introduction

What is Stein's method?
- The usual framework
- The generator approach

The Gaußian case: the OU semi-group
- Behavior of the Stein solution
- The Berry-Esseen bound

Application to extreme value distributions
- Definitions and general results
- Rates of convergence in extreme value theory

Limit theorems in probability

- Limit theorems are ubiquitous in probability theory and its applications.
  - **Examples:** the LLN, the LSN, the CLT, Donsker's theorem, *etc.*
- Play an important role in all fields of mathematics: statistics, functional analysis, potential theory, combinatorics, arithmetic, statistical mechanics, *etc.*
- Proofs rely on powerful tools like *characteristic functions* and *tightness*.
- But how to evaluate the accuracy of those approximations?

Stein's method

- Specific methods for each problem: method of moments, Fourier analysis, *etc.*

Alternative $\longrightarrow$ Stein's method!

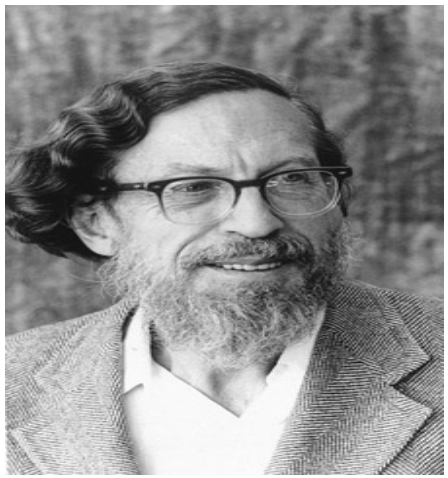


Figure: C. Stein

- Developed in the early 70s by Charles Stein for one of his lectures.

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Probability metrics

Refresher: a sequence of probability measures $(\nu)_{n\geq 1}$ *converges in distribution* to ν if

$$\mathbb{E}_{\nu_n}[h] := \int h \, \mathrm{d}\nu_n \underset{n\to\infty}{\to} \mathbb{E}_{\nu}[h]$$

for all continuous bounded $h : E \to \mathbb{R}$.

- Several ways to metrize this topology: *Lévy-Prokhorov metric*, *Kullback-Leibler divergence*, *etc.*
- Stein's method works for *integral probability metrics*:

$$\mathrm{d}_{\mathcal{H}}(\mu,\nu) = \sup_{h\in\mathcal{H}} \left| \int h \, \mathrm{d}\mu - \int h \, \mathrm{d}\nu \right| = \sup_{h\in\mathcal{H}} |\mathbb{E}_{\mu}[h] - \mathbb{E}_{\nu}[h]|.$$

- **Examples:**
  - If we take

$$\mathcal{H} = \{x \mapsto \mathbf{1}_{(-\infty,z]}(x),\ z \in \mathbb{R}\}$$

    we get the *Kolmogorov distance*:

$$\mathrm{d_K}(\mu,\nu) = \sup_{z\in\mathbb{R}} |\mu(-\infty,z] - \nu(-\infty,z]|.$$

  - Choosing instead

$$\mathcal{H} = \mathrm{Lip}_1(\mathbb{R}) := \{h : \mathbb{R} \to \mathbb{R},\ h \text{ is } 1\text{−Lipschitz}\},$$

    we recover the *Wasserstein distance*:

$$\mathrm{d_W}(\mu,\nu) = \sup_{h\in\mathrm{Lip}_1(\mathbb{R})} (\mathbb{E}_\mu[h] - \mathbb{E}_\nu[h]).$$

  - The *Fortet-Mourier* distance comes with:

$$\mathcal{H} = \{h \in \mathcal{C}(\mathbb{R}),\ \|h\|_\infty \le 1\}.$$

- Requires two ingredients:
  1. finding a *Stein operator* $\mathscr{L}$ for every max-stable distribution ν, *i.e.*

$$\mu = \nu \iff \mathbb{E}_\mu[\mathscr{L}g] = 0 \text{ for all } g \in \mathcal{G},$$

     for some large enough function space $\mathcal{G}$;
  2. solving *Stein equation*, *i.e.* finding $g_h \in \mathcal{G}$ s.t.:

$$\mathscr{L}g_h = h^* := h - \mathbb{E}_\nu[h],\ h \in \mathcal{H}.$$

- Then

$$\mathrm{d}_{\mathcal{H}}(\mu, \nu) \leq \sup_{g \in \mathcal{G}} |\mathbb{E}_\mu[\mathscr{L}g]|.$$

- **Example:** for $\nu = \mathcal{N}(0,1)$, take $\mathscr{L}g(x) = -xg(x) + g'(x)$.

How to construct a characterizing operator $\mathscr{L}$ for a given ν?

Finding Stein operators [Döb12]

Let Z be a real random variable with distribution ν.

- The *differential equation approach*: let A be s.t.

$$Af_Z = 0.$$

Then $\mathscr{L} := A^\dagger$ is a Stein operator:

$$\mathbb{E}[A^\dagger g(Z)] = \int_{\mathbb{R}} A^\dagger g(z) f_Z(z)\, \mathrm{d}z = \int_{\mathbb{R}} g(z) A f_Z(z)\, \mathrm{d}z = 0.$$

Example: The *semi-circle distribution* on $[-1, 1]$

$$f_Z(x) = \frac{2}{\pi}\sqrt{1-x^2} \longrightarrow Af(x) := (1-x^2)f'(x) + xf(x),$$

hence the Stein's operator:

$$\mathscr{L}g(x) = A^\dagger g(x) = (x^2-1)g'(x) + 3xg(x).$$

- By using the derivative of the **log-density** $\psi_Z := f_Z'/f_Z$:

$$\mathbb{E}\big[g'(Z) + g(Z)\psi_Z(Z)\big] = g(b^-)f_Z(b^-) - g(a^+)f_Z(a^+),$$

if Z has support over (a, b). We can take:

$$\mathscr{L}g(x) := g'(x) + g(x)\psi_Z(x) - g(b^-)f_Z(b^-) + g(a^+)f_Z(a^+).$$

Example: The *gamma distribution* on $\mathbb{R}_+$

$$f_Z(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} \longrightarrow \psi_Z(x) = \frac{\alpha - 1 - \lambda x}{x},$$

hence the Stein's operator:

$$\mathscr{L}g(x) = g'(x) + \frac{\alpha - 1 - \lambda x}{x} g(x).$$

The generator approach [Bar88]

- Find a semi-group $(P_t)_{t\geq 0}$ *quantizing* ν:
  1. P_t is linear on $\mathcal{H}$
  2. $P_0 = \mathrm{Id}$
  3. $P_t \circ P_s = P_{t+s}$
  4. $\mathbb{E}_\nu[P_t h] = \mathbb{E}_\nu[h]$
- Then $\mathscr{L}$ is a Stein operator for ν and $\mathcal{H}$, where

$$\mathscr{L} := \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} P_t.$$

- Besides, $\mathscr{L}$ is invertible on $\mathcal{H}_0$ and $g_h = \mathscr{L}^{-1} h^*$ equals:

$$g_h = -\int_0^\infty P_t h \, \mathrm{d}t.$$

Examples

- the *Glauber semi-group*:

$$P_t h(x) := \mathbb{E}\Big[h\big(e^{-t} \circ x + (1 - e^{-t}) \circ Z\big)\Big]$$

with $Z \sim \mathcal{P}(\lambda)$ and $\lambda > 0$, has generator:

$$\mathscr{L} g(x) = -x\big(g(x) - g(x-1)\big) + \lambda\big(g(x+1) - g(x)\big),\ x \in \mathbb{N}.$$

- the α*-stable OU semi-group* [AH23]

$$P_t h(x) := \mathbb{E}\Big[h\big(e^{-t} x + (1 - e^{-\alpha t})^{\frac{1}{\alpha}} Z\big)\Big]$$

where $\phi_Z(x) = e^{-\frac{|x|^\alpha}{2}}$, has generator:

$$\mathscr{L} g(x) = -x g'(x) + \int_{\mathbb{R}} \big(g'(x+u) - g'(x)\big) u \, \mathrm{d}\nu_\alpha(u).$$

Why does it work?

- The *stability postulate* is the key argument:

$$e^{-t}Z + (1 - e^{-\alpha t})^{\frac{1}{\alpha}} Z' \overset{\mathrm{d}}{=} Z.$$

  if Z, Z' are i.i.d. with char. funct. $e^{-\frac{|x|^\alpha}{2}}$, for $\alpha \in (0, 2]$.
- Gives the identity:

$$P_t \circ P_s = P_{t+s}.$$

- A general notion of stability can be defined through the concept of *convex cone* [DMZ08].

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The Ornstein-Uhlenbeck semi-group Nourdin12

Definition (The OU semi-group)

Let $Z \sim \mathcal{N}(0,1)$ and set:

$$P_t h(x) = \mathbb{E}\Big[h\big(e^{-t}x + \sqrt{1-e^{-2t}}Z\big)\Big],$$

for all 1-Lipschitz functions h.

- Markov semi-group of the solution of the EDS:

$$X_t = -X_t \, \mathrm{d}t + \sqrt{2} \, \mathrm{d}B_t. \tag{1}$$

- Equation (1) is also known as the *Langevin equation*.

Generator and Stein's solution

Proposition

$(P_t)_{t\geq 0}$ is a symmetric Markov semi-group admitting $\mathcal{N}(0,1)$ as its invariant measure. Its generator is:

$$\mathscr{L}g(x) = -xg'(x) + g''(x).$$

Proposition

Let $h : \mathbb{R} \to \mathbb{R}$ be a 1-Lipschitz function. The solution g_h to the equation

$$\mathscr{L}g = h^* = h - \mathbb{E}[h(Z)]$$

is given by

$$g_h = -\int_0^\infty P_t h^* \, \mathrm{d}t.$$

This function is well-defined, $\mathcal{C}^2$ and g_h' is also 1-Lipschitz.

Abstract bound for the Wasserstein distance

Define

$$\mathcal{F}_{\mathcal{W}} := \{g \in \mathcal{C}^2(\mathbb{R}),\ \|g''\|_\infty \leq 1\}.$$

Proposition

Let X be a random variable on $\mathbb{R}$ with finite first moment, and $Z \sim \mathcal{N}(0,1)$. Then

$$\mathrm{d_W}\big(\mathcal{L}(X), \mathcal{L}(Z)\big) \leq \sup_{g \in \mathcal{F}_{\mathrm{W}}} \big|\mathbb{E}\big[Xg'(X) - g''(X)\big]\big|. \tag{2}$$

Proof: $\mathbb{E}[h(X)] - \mathbb{E}[h(Z)] = \mathbb{E}\big[h(X) - \mathbb{E}[h(Z)]\big] = \mathbb{E}[\mathscr{L} g_h(X)]$.

The Berry-Esseen bound

Proposition

Let $(X_i)_{i\geq 1}$ be a sequence of i.i.d. centered random variables s.t. $\mathbb{E}[X_1^2] = 1$ and $\mathbb{E}[|X_1|^3|] < \infty$. Then for any $n \geq 1$

$$\mathrm{d_W}(S_n, Z) \leq C\frac{\mathbb{E}[|X_1|^3|]}{\sqrt{n}},$$

with C a universal constant independent of n, $Z \sim \mathcal{N}(0,1)$ and

$$S_n := \frac{1}{\sqrt{n}}\sum_{i=1}^{n} X_i.$$

Outline of the proof (1/3)

We must prove that:

$$\mathbb{E}\big[S_n g'(S_n)\big] \simeq \mathbb{E}[g''(S_n)] + R_n, \tag{3}$$

where R_n can be bounded by $C/\sqrt{n}$. Set

$$X_{n,i} := \frac{1}{\sqrt{n}} X_i,$$

and $S_{n\setminus i} := \sum_{\substack{k=0 \\ k\neq i}}^{n} X_{n,k}$. Then

$$\begin{aligned}\mathbb{E}\big[S_n g'(S_n)\big] &= \sum_{i=1}^{n} \mathbb{E}\big[X_{n,i} g'(S_n)\big] \\ &= \sum_{i=1}^{n} \mathbb{E}\big[X_{n,i} g'(S_{n\setminus i} + X_{n,i})\big].\end{aligned}$$

Outline of the proof (2/3)

We have the Taylor expansion

$$g'(S_{n\setminus i} + X_{n,i}) = g'(S_{n\setminus i}) + X_{n,i} g''(S_{n\setminus i} + \Theta X_{n,i}),$$

where $\Theta \sim \mathcal{U}_{[0,1]}$ is independent of the $X_{n,k}$.

$$\begin{aligned}
\mathbb{E}\big[S_n g'(S_n)\big] &= \sum_{i=1}^{n} \mathbb{E}\big[X_{n,i} g'(S_{n\setminus i} + X_{n,i})\big] \\
&= \sum_{i=1}^{n} \mathbb{E}\big[X_{n,i} g'(S_{n\setminus i})\big] + \mathbb{E}\big[X_{n,i}^2 g''(S_{n\setminus i} + \Theta X_{n,i}))\big] \\
&= \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}\big[X_i^2 g''(S_{n\setminus i} + \Theta X_{n,i})\big].
\end{aligned}$$

Outline of the proof (3/3)

Thus we get

$$
\begin{aligned}
&\left|\mathbb{E}\left[S_n g'(S_n)\right] - \mathbb{E}[g''(S_n)]\right| \\
&\qquad \leq \left|\mathbb{E}\left[S_n g'(S_n)\right] - \mathbb{E}[g''(S_{n\setminus i})]\right| + \left|\mathbb{E}[g''(S_{n\setminus i})] - \mathbb{E}[g''(S_n)]\right| \\
&\qquad \leq \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}\left[X_{n,i}^2 \left|g''(S_{n\setminus i} + \Theta X_{n,i}) - g''(S_{n\setminus i})\right|\right] + \frac{1}{\sqrt{n}}\mathbb{E}[|X_1|] \\
&\qquad \leq \mathbb{E}\left[(1-\Theta)|X_{n,i}|^3\right] + \frac{1}{\sqrt{n}}\mathbb{E}[X_1^2]^{\frac{1}{2}} \\
&\qquad = \frac{1}{\sqrt{n}}\mathbb{E}\left[|X_{n,i}|^3\right] + \frac{1}{\sqrt{n}}.
\end{aligned}
$$

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Univariate extreme value distributions

Here $x \oplus y := \max(x, y)$.

- **Three cases**
  - The *Weibull distribution* $\mathcal{W}(\alpha)$ on $\mathbb{R}_-$:

$$F(x) = e^{-(-x)^\alpha} \longrightarrow aZ \oplus bZ' \stackrel{\mathrm{d}}{=} (a^{-\alpha} + b^{-\alpha})Z.$$

  - The *Gumbel distribution* $\mathcal{G}(0, 1)$ on $\mathbb{R}$:

$$F(x) = e^{-e^{-x}} \longrightarrow (Z + \log a) \oplus (Z' + \log b) \stackrel{\mathrm{d}}{=} Z + \log(a + b).$$

  - The *Fréchet distribution* $\mathcal{F}(\alpha)$ on $\mathbb{R}_+$:

$$F(x) = e^{-\frac{1}{x^\alpha}} \longrightarrow aZ \oplus bZ' \stackrel{\mathrm{d}}{=} (a^\alpha + b^\alpha)Z.$$

The extreme value theorem

- The previous examples are the only possible non-degenerate limits for a renormalized maximum of n i.i.d. random variables $(X_i)_{i\geq 1}$:

$$Z_n := \frac{1}{a_n}\bigoplus_{i=1}^{n}(X_i - b_n) \xrightarrow[n\to\infty]{\mathrm{d}} \begin{cases} \mathcal{W}(\alpha) \\ \mathcal{G}(0,1) \\ \mathcal{F}(\alpha). \end{cases} \tag{4}$$

- Max-stable counterpart of the *generalized central limit theorem*.

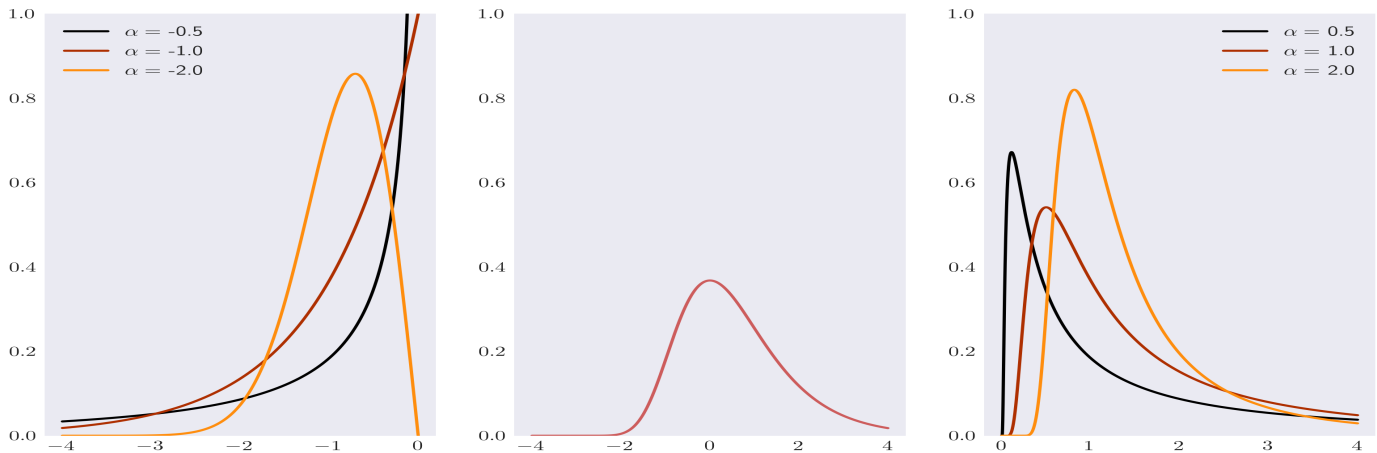


Figure: P.d.f of $\mathcal{W}(\alpha)$, $\mathcal{G}(0,1)$ and $\mathcal{F}(\alpha)$ for $\alpha \in \{0.5, 1, 2\}$

The max-stable semi-groups

We focus on the univariate Fréchet distribution. The next definition is introduced in [CD24b].

Definition (Fréchet semi-group)

Define the *Fréchet semi-group* with index $\alpha > 0$ by

$$\mathbf{P}_t^{\alpha} h(x) := \mathbb{E}\left[h\left(e^{-\frac{t}{\alpha}} x \oplus (1 - e^{-t})^{\frac{1}{\alpha}} Z\right)\right],\ x \geq 0,\ t \geq 0$$

where $Z \sim \mathcal{F}(\alpha)$ and h is a "nice" function.

- That construction is easy to extend to the other univariate EVDs.
- It works also in higher dimensions with *max-stable random vectors*.
- Works also in infinite dimension with *max-stable processes*.

Main properties of $(\mathbf{P}_t^\alpha)_{t\geq 0}$

The four required properties are satisfied:

Proposition

$(\mathbf{P}_t^\alpha)_{t\geq 0}$ is an ergodic semi-group which admits the Fréchet distribution $\mathcal{F}(\alpha)$ as a stationary measure.

Proposition

If h is differentiable, then so is $\mathbf{P}_t^\alpha h$ and

$$(\mathbf{P}_t^\alpha h)'(x) = e^{-\frac{t}{\alpha}} h'\left(e^{-\frac{t}{\alpha}} x\right) e^{-\frac{\gamma_t}{x^\alpha}},$$

where $\gamma_t := e^t - 1$. Therefore:

$$\mathbf{P}_t^\alpha h(x) = -e^{-\frac{t}{\alpha}} \int_x^\infty h'\left(e^{-\frac{t}{\alpha}} r\right) e^{-\frac{\gamma_t}{r^\alpha}} \, \mathrm{d}r.$$

Representation of $(\mathbf{P}_t^\alpha)_{t\geq 0}$ as a Markov process

- There exists a Markov process $(X_t)_{t\geq 0}$ s.t.

$$\mathbf{P}_t^\alpha h(x) = \mathbb{E}\big[h(X_t) \,|\, X_0 = x\big].$$

- That process can be expressed in terms of *extremal integrals* (see [ST05]):

$$X_t := e^{-\frac{t}{\alpha}} X_0 \oplus \int_0^{e^t} e^{-\frac{1}{\alpha}(t-s)} \, \mathrm{d}M_\alpha(s),$$

where M_α is a α-Fréchet random sup-measure with Lebesgue control measure.

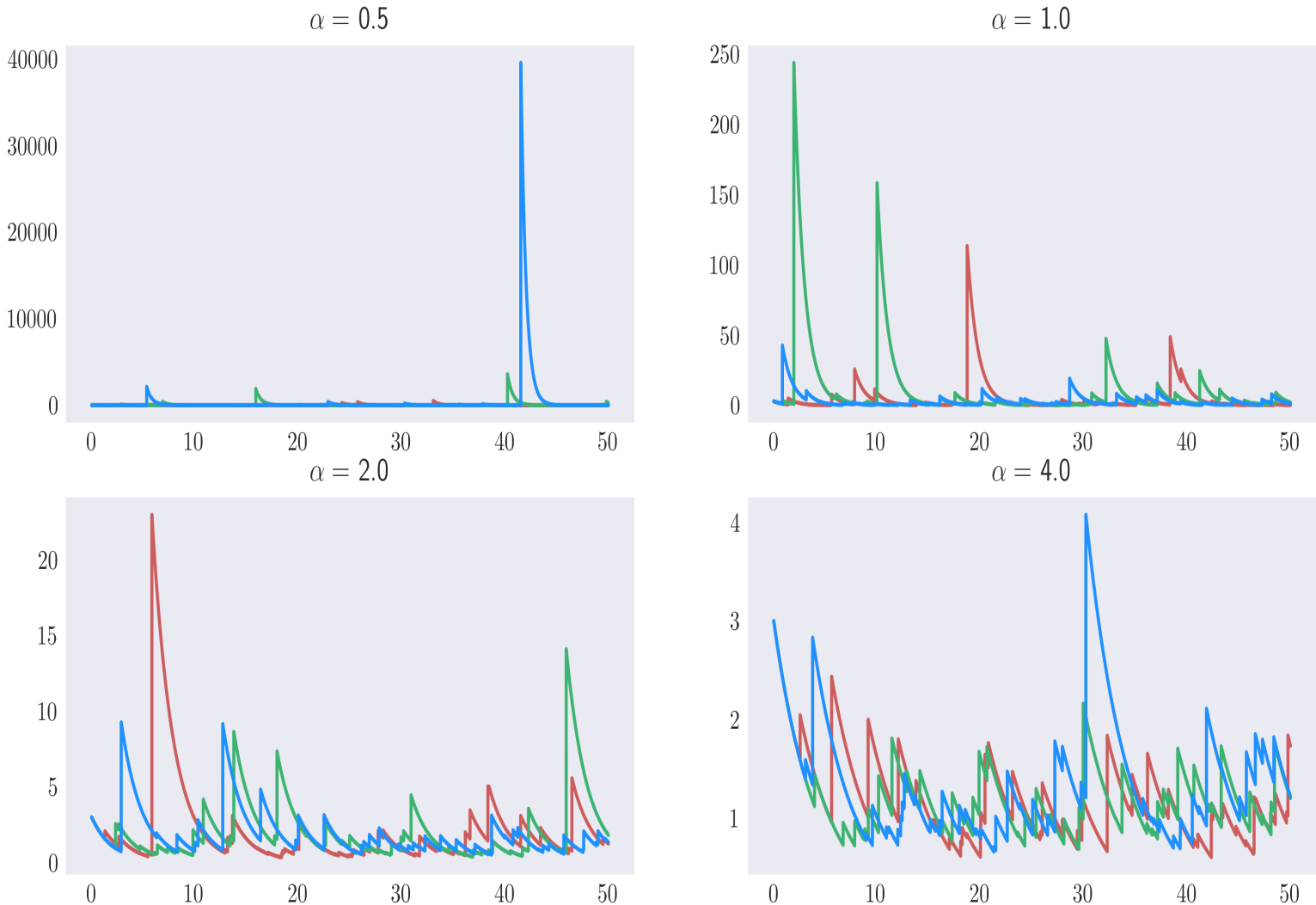


Figure: Simulations of X_t starting at $X_0 = 3$ with $\alpha \in \left\{\frac{1}{2}, 1, 2, 4\right\}$

Main properties of $\mathscr{L}_\alpha$

The generator of the Fréchet semi-group satisfies the following properties:

Proposition

The generator $\mathscr{L}_\alpha$ of $(\mathbf{P}_t^\alpha)_{t\geq 0}$ equals:

$$\mathscr{L}_\alpha g(x) = -\frac{1}{\alpha} x g'(x) + \mathbf{D}_\alpha g(x),$$

with

$$\mathbf{D}_\alpha g(x) := \frac{1}{\alpha}\int_x^\infty r g'(r)\, \mathrm{d}\rho_\alpha(r) = \int_x^\infty g'(r)\frac{\mathrm{d}r}{r^\alpha}.$$

and $\rho_\alpha[x, +\infty) := x^{-\alpha}$.

Proposition (Commutation rule)

$$\mathbf{D}_\alpha \mathbf{P}_t^\alpha = e^{-t}\mathbf{P}_t^\alpha \mathbf{D}_\alpha.$$

The inverse of $\mathscr{L}_\alpha$

Stein's equation and its solution take the following shape:

Proposition (Stein's equation)

Let Z be a positive random variable. Then $X \sim \mathcal{F}(\alpha)$ iff

$$\mathbb{E}[Xg'(X)] = \alpha\mathbb{E}[\mathbf{D}_\alpha g(X)] \tag{5}$$

for all differentiable g s.t.

$$|g'(r)| \leq \frac{C}{r}$$

Proposition (Stein's solution)

1. $\mathscr{L}_\alpha^{-1} h^*$ exist for all $h \in \mathrm{Lip}_1(\mathbb{R}_+, \mathbb{R})$ iif $\alpha > 1$.
2. If $h_z = \mathbf{1}_{(-\infty,z]}$, then $g_z := -\mathscr{L}_\alpha^{-1} h_z^*$ exists and

$$g_z(x) = \alpha(\log xz^{-1})_+ e^{-\frac{1}{z^\alpha}} - \int_{\alpha(\log xz^{-1})_+}^{\infty} \left(e^{-\frac{1-e^{-t}}{z^\alpha}} - e^{-\frac{1}{z^\alpha}} \right) \mathrm{d}t.$$

Refresher

- **Notations:** Let $X_1, \dots, X_n$ be i.i.d. random variables with c.d.f. F_X.
- Assume
$$\overline{F}_X(x) := 1 - F_X(x) \underset{x\to\infty}{\sim} \frac{C}{x^\alpha}.$$
- Set also $a_n := n^{1/\alpha}$ and
$$Z_n := a_n^{-1} \max(X_1, \dots, X_n).$$
- Then
$$Z_n \underset{n\to\infty}{\longrightarrow} \mathcal{F}(\alpha).$$
- **Goal:** Bound the distance between Z_n and $\mathcal{F}(\alpha)$.
- Find a general bound for $\mathrm{d}(W, Z)$ and specialize it to the case $W = Z_n$.

General bound

Proposition

Let W be a random variable such that its c.d.f. F_W is absolutely continuous over $[K, +\infty)$ for some $K > 0$. Set $f_W := F_W'$ and let $Z \sim \mathcal{F}(\alpha)$.

1. If $\alpha > 1$ and $\mathbb{E}[|W|^p] < \infty$ for some $p > 1$, with $p^{-1} + q^{-1} = 1$,

$$d_W(W, Z) \leq \left(C_\alpha + \mathbb{E}\left[|W|^p\right]^{\frac{1}{p}}\right) F_W(K)^{\frac{1}{q}} + \frac{1}{\alpha} \int_K^\infty r \min(\alpha, r^\alpha) F_W(r) \left| \frac{f_W(r)}{F_W(r)} - \frac{\alpha}{r^{\alpha+1}} \right| dr.$$

2. For any $\alpha > 0$,

$$d_K(W, Z) \leq 2F_W(K) + \sup_{z>K} \int_z^\infty e^{-\left(\frac{1}{z^\alpha} - \frac{1}{r^\alpha}\right)} F_W(r) \left| \frac{f_W(r)}{F_W(r)} - \frac{\alpha}{r^{\alpha+1}} \right| dr.$$

Application to Z_n

- Let $\delta, \delta' > 0$ be s.t.

$$\frac{\delta}{r^\alpha} \leq \overline{F}_X(r) \leq \frac{\delta'}{r^\alpha}, \ r \geq K.$$

- Because $F_X \in D(\Phi_\alpha)$, one only needs to assume that

$$\mathbb{E}[X_-^p] < \infty.$$

- If $\alpha > 1$, then we find

$$\mathrm{d_W}(Z_n, Z) \leq C^{\mathrm{W}}_{\alpha, F_X}\left[\frac{1}{n} + n^{1-\frac{1}{\alpha}} \int_K^\infty e^{-\frac{\delta}{2}\frac{n}{r^\alpha}} \left| r^{\alpha+1} f_X(r) - \alpha \right| \frac{\mathrm{d}r}{r^\alpha}\right], \ n \geq 2.$$

- For all $\alpha > 0$,

$$\mathrm{d_K}(Z_n, Z) \leq C^{\mathrm{K}}_{\alpha, F_X}\left[\frac{1}{n} + n \int_K^\infty e^{-\frac{\delta}{2}\frac{n}{r^\alpha}} \left| r^{\alpha+1} f_X(r) - \alpha \right| \frac{\mathrm{d}r}{r^{\alpha+1}}\right], \ n \geq 2.$$

Special case

Corollary

Assume that F_X is a distribution function satisfying the previous assumptions. If its derivative f_X also satisfies

$$|r^{\alpha+1} f_X(r) - \alpha| \lesssim \frac{1}{r^{\gamma}}, \ r \geq K \tag{6}$$

for some $\gamma > 0$, then

- *if $\alpha > 1$:*

$$\mathrm{d_W}(Z_n, Z) \leq C^{\mathrm{W}}_{\alpha, F_X} \left[\frac{1}{n} + \frac{1}{n^{\frac{\gamma}{\alpha}}} \right], \ n \geq 1, \tag{7}$$

- *for all $\alpha > 0$:*

$$\mathrm{d_K}(Z_n, Z) \leq C^{\mathrm{K}}_{\alpha, F_X} \left[\frac{1}{n} + \frac{1}{n^{\frac{\gamma}{\alpha}}} \right], \ n \geq 1. \tag{8}$$

Examples

- If X has the *Pareto distribution with parameter* α, then (6) is trivial:

$$\mathrm{d}(Z_n, Z) \leq \frac{1}{n}.$$

- If X has the *Student distribution with* ν *degrees of freedom*, then

$$f_X(r) = \frac{1}{C}\left(1 + \frac{r^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \ r \in \mathbb{R}.$$

We have for $\sigma := \nu C^{-2/(\nu-1)}$:

$$|r^{\nu+1} f_{\sigma^{-1}X}(r) - \nu| \lesssim \frac{1}{r^2},$$

so that

$$\mathrm{d}(\sigma^{-1} Z_n, Z) \lesssim \frac{1}{n} + \frac{1}{n^{\frac{2}{\nu}}}.$$

Conclusion

- The previous results are readily adapted to the Weibull distribution.
  - Harder to apply to the Gumbel distribution.
- How to generalize to discontinuous distributions, especially to discrete distributions? [CD24a]
- Application to stationary sequences of r.v.?
- Possible extensions to infinite dimension: max-stable processes.
- Max-stable distributions are not the only laws in EVT $\rightarrow$ GPDs.

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